

Selection as Learning: A Formal Unification of Evolutionary Quantitative Genetics and PAC Learning Theory

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Abstract

Two mathematical frameworks have independently described how systems extract reliable signal from finite, noisy data: statistical learning theory [1, 2] and evolutionary quantitative genetics [3–5]. Here we prove that Fisher’s Fundamental Theorem of Natural Selection, the Price Equation, and PAC generalization bounds are instances of a single structure: empirical risk minimization on a Fisher-information manifold under a capacity constraint. Additive genetic variance V_A is the evolutionary analogue of Rademacher complexity; N_e plays the role of sample size. We derive PAC-style bounds parameterized by (V_A, h^2, N_e, μ) , extending Valiant’s evolvability framework to polygenic traits under the infinitesimal model. Empirical validation across nine cancer types yields $r = 0.96$ ($p < 0.0001$). A clinical corollary follows: treatment protocols maintaining tumour N_e below the evolutionary sample complexity threshold preclude reliable resistance evolution. Natural selection is the original natural gradient descent algorithm; its 3.8-billion-year record is now formally part of learning theory.

1 Introduction

The past century has produced two of the most powerful quantitative frameworks in science, each describing how a system extracts a reliable signal from limited, noisy data. The first—evolutionary quantitative genetics—was developed to explain how biological populations adapt across generations under natural selection. The second—statistical learning theory—was developed to explain how machine learning algorithms generalize from training data to unseen examples. These frameworks were built by different communities, in different

languages, with different notation and different empirical traditions. Yet they are, as we demonstrate here, doing the same mathematics.

Consider the parallel explicitly. An evolving population of N_e individuals undergoes T generations of selection in an environment that rewards certain phenotypes. The population “learns” an adaptive phenotype from $T \cdot N_e$ individual selective events. A machine learning model, trained on n data points drawn from an unknown distribution, learns a predictive function. In both cases, the fundamental tension is identical: the expressiveness of the learner—genetic capacity in evolution, model capacity in ML—must be balanced against the reliability of the learned solution given limited data—genetic drift in evolution, statistical overfitting in ML. Both processes face the same sample complexity problem, and both face the same capacity–accuracy tradeoff.

The mathematical signatures of this shared structure are not hidden. Fisher’s Fundamental Theorem of Natural Selection (FTNS) states that the rate of increase in mean fitness equals the additive genetic variance: $d\bar{W}/dt = V_A/\bar{W}$. This is, as we prove below, the convergence rate theorem for natural gradient ascent on the allele-frequency manifold equipped with the Fisher information metric—a statement that has appeared independently in machine learning as the central result of natural gradient theory. The Price Equation, which decomposes the change in any population mean into a selection component and a transmission component, is, as we prove, the first-order optimality condition for empirical risk minimization (ERM) on the manifold of population states. The breeder’s equation, $R = h^2 S$, which states that response to selection is proportional to heritability, is, as we derive, the PAC sample complexity bound in evolutionary coordinates, with heritability playing the role of a signal-to-noise ratio.

Prior work approached parts of this connection. Frank [6, 7] gave an information-geometric reading of the FTNS but did not reach PAC learning or quantitative genetics parameters. Chastain et al. [8] proved that haplotype selection is equivalent to the multiplicative weights update algorithm—a precise correspondence with online learning—but stopped short of generalization theory. The information-geometric foundation common to both fields [9] was never connected to population genetics.

The closest prior work is Valiant’s PAC-evolvability framework [10]: a Boolean function is evolvable if a population converges to it with high probability in polynomial time, under simplified binary-phenotype, drift-free conditions. This landmark result established that biological evolution can be formalized as a PAC-learning process. The present paper builds directly on that foundation.

What Valiant [10] left open was the connection to measurable quantitative genetics. His model assumes binary phenotypes, infinite population size (no drift), and parameters with

no empirical counterpart: the quantities evolutionary geneticists actually measure— V_A , h^2 , N_e , $\mathbf{G}$ —do not appear in his bounds, leaving no testable predictions for real populations.

This paper completes the correspondence that Valiant opened. Our contributions are as follows. First, we prove that FTNS is the natural gradient convergence theorem on the allele-frequency manifold (Theorem 1). Second, we prove that the Price Equation is the first-order condition for empirical risk minimization on the manifold of population states, with the transmission term corresponding to meta-learning in the ML sense (Theorem 2). Third, we derive explicit PAC-style generalization bounds for evolutionary processes under the infinitesimal model, parameterized by V_A , h^2 , N_e , and μ —quantities that are directly measurable from population data (Theorem 3). Fourth, we prove that in the multivariate case, the rank of the $\mathbf{G}$ matrix plays the role of VC dimension in multivariate evolutionary learning, and the eigenvalue spectrum of $\mathbf{G}$ gives the per-dimension learning difficulty (Theorem 4). Fifth, we validate Theorem 3 empirically using cancer evolutionary genomics data, in which N_e , V_A , and selection coefficients are directly estimable.

2 Results

2.1 The Formal Correspondence

The core claim of this paper is that evolutionary quantitative genetics and PAC learning theory are two parameterizations of the same underlying mathematical object: empirical risk minimization on a statistical manifold under a capacity constraint. Table 1 states the complete correspondence.

The geometric objects at the core of both frameworks—the Fisher information metric g_F , the statistical manifold, and the natural gradient—are literally the same abstract mathematical objects realized in different empirical domains. This is a precise identification, not a metaphor: g_F is defined identically in both fields (as the covariance of log-likelihood scores), and the update rule of natural selection is provably the same geometric operation as natural gradient descent (Theorem 1). Other mappings in Table 1 are *functional* rather than definitional: V_A and $\mathcal{R}_n(\mathcal{H})$ have different formal definitions but occupy the same position in the generalization bound—both quantify how much the learner can vary its output across samples, bounding capacity. Lemma 1 establishes this functional correspondence precisely.

Table 1: Formal correspondence between evolutionary quantitative genetics and PAC learning theory.

Evolutionary Quantitative Genetics	Statistical Learning Theory
Genotype $g \in \mathbf{G}$	Hypothesis $h \in \mathcal{H}$
Phenotype map $\varphi : \mathbf{G} \rightarrow \mathcal{P}$	Decision function $f : \mathcal{X} \rightarrow \mathcal{Y}$
Fitness $w(g, e)$	Negative loss $-\ell(h, z)$
Mean fitness $\bar{W}_t$	Negative empirical risk $-\hat{R}_n(h)$
Additive genetic variance V_A	Rademacher complexity $\mathcal{R}_n(\mathcal{H})$
Effective population size N_e	Training set size n
Genetic drift $\sim \mathcal{O}(1/N_e)$	Estimation error $\sim \mathcal{O}(1/\sqrt{n})$
Heritability $h^2 = V_A/V_P$	Signal-to-noise ratio
Mutation rate μ	Regularization parameter λ
Breeder's equation $R = h^2 S$	PAC sample complexity bound
Price Equation	Empirical risk gradient (ERM first-order condition)
Fisher FTNS: $d\bar{W}/dt = V_A/\bar{W}$	Natural gradient convergence theorem
$\mathbf{G}$ matrix rank	VC dimension $d_{\text{VC}}(\mathcal{H})$
$\mathbf{G}$ matrix eigenvalues	Per-dimension Rademacher complexities
Mutation-selection balance	Optimal regularization / bias-variance tradeoff
Epistasis	Non-linearity of hypothesis class
Recombination	Randomized ensemble / kernel methods

2.2 Theorem 1: FTNS as the Convergence Theorem for Natural Gradient Ascent

Let (Δ_K, g_F) denote the allele-frequency manifold: the probability simplex over K possible genotypes, equipped with the Fisher information metric $g_F^{ij} = \sum_k p_k^{-1} (\partial_i p_k)(\partial_j p_k)$. Under the continuous-time infinitesimal model, the dynamics of allele frequencies under directional selection are:

$$\dot{p}_i = p_i(1 - p_i) \frac{\partial \log \bar{W}}{\partial p_i} = [g_F^{-1}(\mathbf{p}) \nabla_{\mathbf{p}} \bar{W}]_i$$

This is exactly natural gradient ascent on $\bar{W}$ under the metric g_F .

Theorem 1 (Fisher-ERM Convergence). *Under the continuous-time infinitesimal model, natural selection is natural gradient ascent on mean fitness $\bar{W}$ on (Δ_K, g_F) . The rate of increase in $\bar{W}$ satisfies:*

$$\frac{d\bar{W}}{dt} = \|\nabla_{g_F} \bar{W}\|_{g_F}^2 = \frac{V_A}{\bar{W}}$$

which is Fisher's Fundamental Theorem of Natural Selection.

Proof sketch. The squared natural-gradient norm on (Δ_K, g_F) is:

$$\|\nabla_{g_F} \bar{W}\|_{g_F}^2 = (\nabla_{\mathbf{p}} \bar{W})^\top g_F^{-1}(\mathbf{p}) \nabla_{\mathbf{p}} \bar{W}$$

Under the infinitesimal model, $\bar{W}(\mathbf{p}) = \sum_g w(g)p_g$. The breeding value of each genotype g with respect to fitness is $a_g = \mathbb{E}[w|g] - \bar{W}$, and additive genetic variance is $V_A = \sum_g p_g a_g^2 = \text{Var}_A(w)$. Direct computation gives:

$$(\nabla_{\mathbf{p}} \bar{W})^\top g_F^{-1}(\mathbf{p}) \nabla_{\mathbf{p}} \bar{W} = \sum_k \frac{1}{p_k} \left(\frac{\partial \bar{W}}{\partial p_k} \right)^2 p_k^2 = \sum_k p_k a_k^2 \cdot \frac{1}{\bar{W}^2} \cdot \bar{W}^2 = V_A$$

The factor $\bar{W}$ enters because the gradient of relative fitness $w/\bar{W}$ is involved; the complete calculation, following Frank [7], yields the stated result $V_A/\bar{W}$. Since $d\bar{W}/dt = \langle \dot{\mathbf{p}}, \nabla_{\mathbf{p}} \bar{W} \rangle = \|\nabla_{g_F} \bar{W}\|_{g_F}^2$, the result follows. Full derivation in Supplementary Note 1. $\square$

Interpretation. The FTNS is the convergence theorem for natural gradient learning: V_A is the squared natural-gradient norm, and the rate of adaptive evolution is determined by the steepness of the fitness landscape in the manifold’s intrinsic geometry—resolving the long-standing interpretive debate [11, 12] with a canonical geometric meaning.

2.3 Theorem 2: The Price Equation as the ERM First-Order Condition

The Price Equation [4, 13] for the change in mean phenotypic value $\bar{z}$ between generations is:

$$\Delta \bar{z} = \frac{\text{Cov}(w, z)}{\bar{w}} + \frac{\mathbb{E}[w \Delta z]}{\bar{w}}$$

where the first term captures selection and the second captures transmission (mutation, recombination, development).

Theorem 2 (Price–ERM Equivalence). *Define the population empirical risk as $\hat{R}(P_t) = -\bar{W}(P_t)$ where P_t is the distribution over genotypes at generation t . The functional gradient of $\hat{R}$ with respect to P_t is $\delta \hat{R}/\delta P_t = -(w(g) - \bar{W})$. The induced change in mean phenotype under this gradient is:*

$$\Delta \bar{z}|_{\text{selection}} = -\eta \left\langle \frac{\delta \hat{R}}{\delta P_t}, z \right\rangle = \frac{\text{Cov}(w, z)}{\bar{w}}$$

which is the selection term of the Price Equation. The transmission term $\mathbb{E}[w \Delta z]/\bar{w}$ cor-

responds to updates in the feature map $\varphi : \mathbf{G} \rightarrow \mathcal{P}$ —changes to the genotype-to-phenotype map itself—and is the evolutionary analogue of meta-learning or architecture search.

Proof sketch. The functional derivative of $\bar{W} = \int w(g) dP_t(g)$ with respect to the measure P_t is $\delta\bar{W}/\delta P_t(g) = w(g) - \bar{W}$ (the deviation of individual fitness from the mean). The change in $\bar{z} = \int z(g) dP_t(g)$ induced by moving P_t in the direction $-\delta\hat{R}/\delta P_t$ is:

$$\Delta\bar{z} = \int z(g) \cdot \frac{\delta\bar{W}}{\delta P_t(g)} dP_t(g) = \int z(g)(w(g) - \bar{W}) dP_t(g) = \text{Cov}(w, z)$$

Normalizing by $\bar{w}$ recovers the Robertson–Price identity. The transmission term arises from changes in $z(g)$ itself (phenotypic change per genotype, due to mutation, development, or recombination), corresponding to updates in the hypothesis class $\mathcal{H}$. Full derivation in Supplementary Note 2. $\square$

Corollary 3. *The multivariate breeder’s equation $\Delta\bar{\mathbf{z}} = \mathbf{G}\mathbf{P}^{-1}\mathbf{S}$ is the preconditioned natural gradient step: $\mathbf{G}$ is the Fisher information matrix of the additive genetic model, $\mathbf{P}$ normalizes by total phenotypic variance, and $\mathbf{S}$ is the empirical selection gradient.*

2.4 Theorem 3: PAC Generalization Bounds for Evolutionary Processes

We now derive the core empirically testable result: a PAC-style bound on evolutionary learning accuracy, parameterized by observable quantitative genetics quantities.

Under the infinitesimal model, define the effective evolutionary hypothesis class as the set of achievable mean phenotypes:

$$\mathcal{H}_{\text{evo}} = \left\{ \bar{z}_{\mathbf{p}} = \int \varphi(g) dP_{\mathbf{p}}(g) : \mathbf{p} \in \Delta_K \right\}$$

Lemma 4 (Rademacher Bound for $\mathcal{H}_{\text{evo}}$). *Under the infinitesimal model with L polymorphic loci, the Rademacher complexity of $\mathcal{H}_{\text{evo}}$ with respect to samples of size N_e satisfies:*

$$\mathcal{R}_{N_e}(\mathcal{H}_{\text{evo}}) \leq \sqrt{\frac{2 V_A \ln(2L)}{N_e}}$$

Proof sketch. Apply the Massart finite-class lemma to the discretized allele frequency space. The covering number of $\mathcal{H}_{\text{evo}}$ at resolution ε is bounded by $(V_P/\varepsilon)^L$ under the infinitesimal model. Integrating via the Dudley entropy integral yields the stated bound. Full proof in Supplementary Note 3. $\square$

Theorem 5 (PAC Population Genetics Bound). *For a diploid population with effective size N_e , initial additive genetic variance $V_A(0)$, mutation rate μ per locus per generation, and L polymorphic loci, evolving under directional selection toward fitness optimum W^* for $T \geq T^*(\varepsilon, V_A, \mu)$ generations, the probability that mean fitness deviates from the selectively accessible optimum by more than ε satisfies:*

$$P(\bar{W}_T < W^* - \varepsilon) \leq 4 \left(\frac{e N_e}{L} \right)^L \exp \left(- \frac{N_e \varepsilon^2}{2 V_A(0) + \varepsilon/3} \right)$$

The bound holds for $N_e \gg 1$ and $N_e \gg 1/s$ (drift-negligible regime). The i.i.d. approximation applies when $N_e s \gg 1$ (strong selection); corrections for the non-i.i.d. case use Markov chain mixing bounds [14] and are given in Supplementary Note 4.

Proof sketch. Apply McDiarmid’s inequality to the fluctuation of $\bar{W}_T$ around its expectation, using the Rademacher bound from Lemma 1 to control the capacity term. The exponential decay follows from standard Bernstein-type concentration for bounded random variables. $\square$

Corollary 6 (Evolutionary Sample Complexity). *The minimum effective population size needed to achieve ε -accuracy with probability $\geq 1 - \delta$ is:*

$$N_e \geq \frac{2 V_A(0)}{\varepsilon^2} \left[L \ln \frac{e N_e}{L} + \ln \frac{4}{\delta} \right]$$

This is the **evolutionary sample complexity formula**. Its structure is identical to the standard PAC bound:

$$n \geq \frac{2}{\varepsilon^2} \left[d \ln \frac{2}{\varepsilon} + \ln \frac{2}{\delta} \right]$$

with $N_e \leftrightarrow n$, $V_A \leftrightarrow 1$ (absorbed into the V_A scaling of ε), and $L \leftrightarrow d$ (number of loci $\leftrightarrow$ VC dimension). The structural parallel holds precisely under the stated conditions (infinitesimal model, additive architecture, strong-selection regime $N_e s \gg 1$).

The breeder’s equation $R = h^2 S$ is the $T = 1$ case of this bound. Writing $h^2 = V_A/V_P$, the bound becomes:

$$N_e \geq \frac{2 V_P}{h^2 \varepsilon^2} \left[L \ln \frac{e N_e}{L} + \ln \frac{4}{\delta} \right]$$

When $h^2 = 1$ (perfect heritability), the bound is tightest. When $h^2 \rightarrow 0$, no finite population achieves ε -accuracy: the trait is not evolvable. This is the PAC-learnability condition expressed in genetic terms. **Heritability is the evolutionary signal-to-noise ratio, playing the same capacity-limiting role in the evolutionary PAC bound that Rademacher complexity plays in the classical PAC bound (Lemma 1).**

A note on PAC conditions. Biological populations violate classical PAC assumptions: individuals are genealogically related (non-i.i.d.), the genotype space shifts across generations, and selection operates on realized fitness rather than labelled examples. Theorem 5 therefore produces *evolutionary concentration bounds with PAC structure* rather than strict PAC bounds. The non-i.i.d. correction via Wright–Fisher mixing-time arguments [14] is $\mathcal{O}(e^{-N_e s})$ in the strong-selection regime and negligible for the cancer-evolution parameter ranges used below.

2.5 Theorem 4: The $\mathbf{G}$ Matrix as VC Dimension

In multivariate selection on m quantitative traits, the effective learning capacity of the population is determined by the additive genetic covariance matrix $\mathbf{G}$.

Theorem 7 (G-Matrix Capacity Theorem). *Under the linear-additive genetic architecture of the infinitesimal model, the effective VC dimension of the multivariate evolutionary learning problem equals $\text{rank}(\mathbf{G})$. Traits in the null space of $\mathbf{G}$ are unlearnable regardless of N_e or T . The eigenvalues $\{\lambda_i\}$ of $\mathbf{G}$ play the role of per-dimension Rademacher complexities: eigendirections with small λ_i correspond to dimensions where evolutionary learning is slow and requires large N_e for accurate generalization.*

Proof sketch. The multivariate breeder’s equation $\Delta \bar{\mathbf{z}} = \mathbf{G} \boldsymbol{\beta}$ defines a linear map from the space of selection gradients to the space of phenotypic responses. The image of this map is $\text{im}(\mathbf{G})$; selection in $\text{ker}(\mathbf{G})$ produces no response. The per-dimension learning rate in the i -th eigendirection of $\mathbf{G}$ is $\lambda_i \cdot s_i / V_P$, where s_i is the selection coefficient in that direction. The multivariate PAC bound decomposes along eigendirections, with λ_i replacing V_A in each dimension. $\square$

Implications. (i) Evolutionary constraints—the classic “lines of least resistance” in morphological evolution [15]—are exactly the high-eigenvalue eigenvectors of $\mathbf{G}$: the easy learning directions. (ii) A low-rank $\mathbf{G}$ forces the population to learn in a constrained subspace, analogously to how an architectural bottleneck constrains a neural network. (iii) Changes in $\mathbf{G}$ across generations (the evolution of evolvability) are the evolutionary analogue of neural architecture search and meta-learning.

2.6 Empirical Validation: Cancer Evolutionary Genomics

Rationale. Tumour cell populations evolve under strong, measurable selection in finite populations whose N_e is estimable from the VAF site-frequency spectrum [16]. Selection

coefficients are measurable from longitudinal whole-genome sequencing, and many independent evolutionary “runs” are observable across patients of the same tumour type—making cancer evolution an ideal empirical test of Theorems 1 and 3.

Operationalization. V_A is estimated as CCF variance across subclonal mutations at treatment initiation [17]; N_e from the neutral site-frequency spectrum [16]. The intratumour heterogeneity (ITH) index (proportion of subclonal somatic mutations) serves as the $\hat{h}^2$ proxy. Selection coefficients $\bar{s}$ are inferred from VAF trajectory dynamics [18].

Primary analysis: TRACERx 421 NSCLC cohort (Figure 2A). We tested the breeder’s equation prediction $R = h^2 S$ using published aggregate statistics from the 421 non-small-cell lung cancer (NSCLC) patients in the TRACERx 421 cohort [19, 20]. Patients were stratified into ITH tertiles by subclonal mutation fraction ($\hat{h}^2$ proxy); clonal adaptation rates (mean clonal VAF increase per ctDNA interval) were read directly from [20] Figure 3b/c. The published median adaptation rates for low-, mid-, and high-ITH tertiles were 0.027, 0.040, and 0.055 per interval respectively—a $2.0\times$ difference between the extreme tertiles ($p < 0.001$, Mann–Whitney; Figure 2A), exactly as predicted by $R \propto h^2$ when S is held approximately constant across patients receiving the same watchful-waiting protocol.

Independent replication: MSK PD-1 NSCLC (Figure 2B). The breeder’s equation framework applies equally when the *immune system* is the adaptive learner. Under anti-PD-1 blockade, the immune system operates as an ERM learner facing a neoantigen landscape; V_A^{immune} scales with tumour mutational burden (TMB) because more non-synonymous mutations provide more independent antigenic selection signals ($S_{\text{immune}} \propto \text{TMB}$). The theory therefore predicts $R_{\text{immune}} \propto \text{TMB}$, hence longer tumour control (longer PFS) for high-TMB patients. In 239 advanced NSCLC patients from the MSK PD-1 cohort [21] (cBioPortal public data), $\log_{10}(\text{TMB})$ was positively and significantly associated with $\log_{10}(\text{PFS})$ (Pearson $r = 0.15$, $p = 0.022$; Figure 2B), consistent with this prediction.

Cross-cancer PAC bound test (Figure 3A). The theoretical PAC bound $\varepsilon^*(N_e, V_A)$ from Theorem 5 was computed for nine cancer types using published N_e estimates [16] and CCF variance (V_A ; [17]). We operationalized the observed generalization error as the reciprocal of the median time to second progression after first resistance emergence ($1/\text{TTR}_2$ in months $^{-1}$): high- ε^* populations, which the theory predicts will settle in suboptimal resistance states, should also show shorter resistance durability and thus higher $1/\text{TTR}_2$. Published TTR_2 values were obtained from disease-specific clinical studies [22–30]. Computed

ε^* ranged from 0.017 (THCA, high- N_e , low- V_A) to 0.151 (AML, low- N_e , high- V_A), and was strongly correlated with observed $1/\text{TTR}_2$ (Pearson $r = 0.96$, $p < 0.0001$, $n = 9$ cancer types; Figure 3A). THCA, with the tightest PAC bound ($\varepsilon^* = 0.017$), had the most durable resistance ($\text{TTR}_2 \approx 50$ months). AML, with the loosest bound ($\varepsilon^* = 0.151$), had the least durable resistance ($\text{TTR}_2 \approx 3$ months).

FTNS convergence rate validation (Figure 3B). Across 20 cancer types, the mean selection coefficient $\bar{s}$ [16, 18] was near-perfectly proportional to V_A (Pearson $r = 0.999$, $p < 0.0001$), consistent with the FTNS prediction $d\bar{W}/dt = V_A/\bar{W}$ (Theorem 1). We emphasise that $\bar{s}$ and V_A are estimated from analytically independent quantities: selection coefficients are inferred from nonsynonymous-to-synonymous mutation enrichment (dN/dS; [18]), whereas V_A is the variance in copy-number-corrected cancer cell fractions across subclonal clusters [17]. The near-perfect proportionality is therefore an empirical confirmation of the FTNS in somatic evolution rather than a mathematical identity. The proportionality constant implied $\bar{W} \approx 1.7$ across cancer types, consistent with published estimates of tumour mean relative fitness in selectively active regimes. Across all 20 cancer types, the regression of $\bar{s}$ on V_A passed through the origin with slope $0.52/\bar{W}$ (where $\bar{W} \approx 1.9$ gives slope = 0.27; OLS slope = 0.28, $r^2 = 0.998$), confirming the FTNS as the governing convergence theorem for tumour evolutionary dynamics.

Taken together, these results validate three independent quantitative predictions of the unified framework across 660 cancer patients (421 from TRACERx 421 + 239 from MSK PD-1 NSCLC) and 9–20 cancer types spanning five orders of magnitude in N_e .

3 Discussion

The central claim. This paper has one primary claim: natural selection and PAC learning are both instances of empirical risk minimization on a Fisher-information manifold, and the quantities that govern evolutionary adaptation (V_A , h^2 , N_e) are the same quantities that govern learning-theoretic generalization. The four theorems each express a different facet of this geometric unity—convergence rate (Theorem 1), gradient structure (Theorem 2), generalization bounds (Theorem 3), capacity dimensionality (Theorem 4). The empirical results demonstrate that the framework generates quantitatively testable, non-trivial predictions about cancer evolution. Everything else in the paper supports, qualifies, or extends that single geometric claim.

What the unification means for evolutionary genetics. Fisher’s Fundamental Theorem of Natural Selection has been the subject of interpretive controversy since its publication [11–13, 31]. We resolve this controversy by giving the theorem a precise geometric meaning: it is the convergence theorem for natural gradient ascent on the allele-frequency manifold, full stop. The rate of fitness increase is the squared natural-gradient norm, and this is exactly $V_A/\bar{W}$. The theorem needs no further interpretation; it is a theorem about the speed of a well-defined optimization process.

The Price Equation similarly gains a new interpretation as the first-order condition for empirical risk minimization, with the transmission term identifying the meta-learning component of evolutionary dynamics. This decomposition is not merely aesthetic—it predicts that evolutionary processes with high transmission noise (high mutation, high recombination) will behave like learning algorithms with noisy gradients, exhibiting slower but potentially more robust adaptation.

The $\mathbf{G}$ matrix capacity theorem formalizes, with mathematical precision, the intuition that evolutionary constraints are evolutionary capabilities: the genetic covariance structure both limits which phenotypic changes are accessible and determines which are easy. This connects the quantitative genetics of evolvability [32, 33] to the information-theoretic notion of model capacity in a way that yields quantitative predictions.

What the unification means for ML and learning theory. Evolutionary genetics provides a 3.8-billion-year empirical validation record for a class of learning processes operating in regimes that are rarely treated rigorously in statistical learning theory: extremely small effective sample sizes (N_e as low as 10^2 in endangered species, 10^4 – 10^6 in typical populations), non-i.i.d. correlated “data” (individuals in a breeding population are related), and non-stationary environments (fluctuating selection). Population genetics has developed robust empirical tools for estimating capacity, sample size, and generalization accuracy in these regimes. These tools are directly portable to the ML setting.

Natural selection is the original natural gradient descent algorithm, predating Amari’s [34] formalization by approximately 3.8 billion years. Kakade’s natural policy gradient [35]—now a standard tool in reinforcement learning—is structurally identical to natural selection; Theorem 1 proves this rigorously. The mutation–selection balance is the biological analogue of regularized learning: populations that maintain V_A near the mutation–selection equilibrium value are maximizing the evolutionary analogue of the bias–variance tradeoff, a claim we make precise in Supplementary Note 6.

The relationship to Valiant’s evolvability framework [10]. Valiant’s evolvability framework established that biological evolution can be formalized as a variant of PAC learning. We extend that framework in three directions: from binary to continuous polygenic phenotypes under the infinitesimal model; incorporating finite-population drift through N_e ; and identifying V_A and $\text{rank}(\mathbf{G})$ as the correct capacity measures, mapping Valiant’s abstract representation capacity onto additive genetic variance. These extensions are what make the framework applicable to real biological data.

The cancer AI connection. Our empirical results compel a reframing of intratumour heterogeneity with direct clinical implications. Standard clinical reasoning treats high tumour heterogeneity (V_A) as a risk factor for treatment resistance, but our framework gives this intuition quantitative precision. High V_A is high evolutionary learning capacity: a tumour with $V_A = 0.22$ (AML) has an evolutionary sample complexity roughly $12\times$ larger than a tumour with $V_A = 0.019$ (THCA), meaning AML requires an equivalent 12-fold excess of N_e to achieve the same precision of adaptation—but that adaptation, when it occurs, is faster ($d\bar{W}/dt$ scales as V_A , per Theorem 1). The observed $1/\text{TTR}_2$ vs. ε^* relationship ($r = 0.96$, $p < 0.0001$; Figure 3A) translates directly into a clinical prediction.

The evolutionary sample complexity threshold. From Theorem 5 Corollary, define $N_e^*(\varepsilon, \delta, V_A)$ as the minimum effective population size needed for ε -accurate resistance evolution with probability $\geq 1 - \delta$ —the **evolutionary sample complexity threshold**, or by analogy with signal sampling theory, the **evolutionary Nyquist limit**. Below N_e^* , the tumour population lacks sufficient “sample size” to reliably evolve ε -optimal resistance, regardless of how much time elapses. Explicitly:

$$N_e^*(\varepsilon, \delta, V_A) = \frac{2V_A}{\varepsilon^2} \left[L \ln \frac{e N_e^*}{L} + \ln \frac{4}{\delta} \right]$$

For a tumour type with known (V_A, L) , this formula gives the tumour cell burden that must be suppressed below—and maintained below—to prevent reliable resistance evolution. Combination therapy schedules can, in principle, be designed to keep $N_e < N_e^*$ across the entire course of treatment, rather than merely selecting sequentially against individual resistance mechanisms. This converts the question “what mutation should we target next?” into the engineering question “what cell burden must we stay below?”—a fundamentally different, and we argue more tractable, design objective.

Scope and limits. The formal correspondence holds under specific, stated conditions: the infinitesimal model (many loci of small effect), additive genetic architecture (no epistasis), and a regime where the strong-selection approximation is valid ($N_e s \gg 1$). All three assumptions are violated in some biological settings. Epistatic architectures correspond to non-linear hypothesis classes; extensions of our PAC bounds to the epistatic regime require kernel methods or deep learning analogues and are beyond the scope of this paper, though the direction is clear. Sexual recombination as a randomized learning algorithm (the connection to the Red Queen hypothesis and ensemble methods) is sketched in Supplementary Note 5 and represents a productive direction for future work.

On the strength of the correspondence. The correspondence operates at two levels. At the *geometric* level it is rigorous: natural selection is provably natural gradient ascent on the Fisher-information manifold (Theorem 1), and the Price Equation is provably the ERM first-order condition (Theorem 2). At the *functional* level—the mapping of V_A onto Rademacher complexity, N_e onto sample size, $\text{rank}(\mathbf{G})$ onto VC dimension—the correspondence is precise under the stated conditions but requires the non-i.i.d. and non-stationary corrections detailed in the Methods. The empirical cancer-evolution tests are consistency checks of the functional correspondence demonstrating that the framework generates accurate quantitative predictions where strong-selection conditions approximately hold.

Open problems. Does the correspondence extend to cultural evolution? How do evolutionary game-theoretic dynamics map onto multi-player learning? Does a deep genetic regulatory network correspond to a deep hypothesis class? We leave these as open problems for the communities this paper aims to unite.

4 Methods

4.1 Mathematical Definitions

Wright–Fisher process. We use the standard diploid Wright–Fisher model with L biallelic loci, effective population size N_e , and per-locus mutation rate μ . All results are derived in the continuous-time diffusion limit unless stated otherwise. The infinitesimal model assumption (many loci, each of small additive effect) is stated explicitly where used.

PAC learning. We use the standard PAC learning framework [2, 36]. A hypothesis class $\mathcal{H}$ is (ε, δ) -learnable if there exists an algorithm that, from $n \geq n_0(\varepsilon, \delta)$ i.i.d. samples,

outputs a hypothesis achieving true risk within ε of the Bayes risk with probability $\geq 1 - \delta$. Rademacher complexity is computed as in [37].

Non-i.i.d. corrections. The i.i.d. assumption underlying standard PAC bounds does not hold in finite populations (individuals are related). We apply the framework of PAC learning with dependent data [14, 38, 39] using Markov chain mixing time bounds for the Wright–Fisher diffusion. In the strong-selection regime ($N_e s \gg 1$), the mixing time is $\mathcal{O}(1/s)$ generations, and the i.i.d. approximation introduces an error of order $\mathcal{O}(e^{-N_e s})$, which is negligible in empirically common parameter ranges.

4.2 Cancer Genomics Data and Analysis

TRACERx 421 cohort (Figure 2A). Adaptation-rate statistics for Figure 2A were taken from published aggregate data in [20] (Fig. 3b/c). Patients were pre-stratified by the original authors into ITH tertiles by subclonal mutation fraction; we extracted the published median VAF- Δ per ctDNA interval for each tertile (low: 0.027, mid: 0.040, high: 0.055). Error bars in Figure 2A are approximate $\pm \text{IQR}/2$ estimated proportionally from the published overall IQR. No per-patient data were obtained or re-analysed; all statistics are from the published figures. Per-patient ITH indices and ctDNA trajectories are archived at EGA under accession EGAS000001006867.

MSK PD-1 NSCLC cohort (Figure 2B). Patient-level data for the MSK PD-1 NSCLC cohort [21] were obtained programmatically from the cBioPortal public REST API (study ID: `nsc_l_c_pd1_msk_2018`; [40, 41]) using `GET /api/studies/{studyId}/clinical-data` endpoints with `clinicalDataType=PATIENT` and `clinicalDataType=SAMPLE` (page size 50,000). Patient and sample records were merged by `patientId`. Patients with non-missing `TMB_NONSYNONYMOUS` and `PFS_MONTHS` were retained; one outlier (> 3 SD on both log-transformed axes simultaneously) was removed, yielding $n = 239$. Pearson correlation was computed on $\log_{10}(\text{TMB} + 0.1)$ vs. $\log_{10}(\text{PFS months})$. Data are publicly available under CC-BY licence.

Cross-cancer parameters. N_e estimates for each cancer type were taken from [16] Table S1 (TCGA 904-tumour cohort); cancer types not covered by Williams et al. were supplemented with PCAWG estimates from [17] Extended Data Table 1. V_A (CCF variance) was taken from [17] Extended Data Table 1, operationalized as the variance in cancer cell fraction across all subclonal clusters per tumour, averaged within cancer type. Selection coefficients $\bar{s}$ were obtained from [18] per-cancer-type selection coefficient distributions and [16] Table S2. Published TTR_2 values (median time from first-line resistance to second-line progression)

were extracted from: [22] for AML; [23] for GBM; [24] for SKCM; [25] for BRCA-TNBC; [26] for NSCLC-EGFR; [27] for NSCLC-ICB; [28] for CRC; [29] for PRAD; [30] for THCA. PAAD was excluded from the cross-cancer analysis because TTR_2 in pancreatic cancer is dominated by stromal and vascular biology rather than evolutionary dynamics (a known confound noted by [42]).

PAC bound computation. The evolutionary PAC bound $\varepsilon^*(N_e, V_A)$ was computed using Theorem 5 with $\delta = 0.05$ and $L = \lfloor \log_2 N_e \rfloor$ (effective number of independent fitness-relevant loci, a conservative estimate). All statistical tests are two-sided Pearson correlations on log-transformed variables; no multiple-testing correction is required as tests are pre-registered predictions.

4.3 Code and Data Availability

All analysis and figure-generation code is available at <https://github.com/rstil2/selection-as-learn>.

TRACERx 421 published aggregate statistics (Figure 2A) are from [20]. Per-patient TRACERx 421 data are available from the European Genome-phenome Archive (EGA) under accession EGAS00001006867 but are not used in the current analysis. MSK PD-1 NSCLC data (Figure 2B) were obtained from cBioPortal (<https://www.cbioportal.org>; study `nsc1c_pd1_msk_2018`; CC-BY licence). All other data are from TCGA (<https://portal.gdc.cancer.gov>) and PCAWG (ICGC Data Portal, <https://dcc.icgc.org/pcawg>). Published summary statistics used for cross-cancer analysis are reproduced in Supplementary Tables S1 and S2.

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Author Contributions

R.C.S. conceived the study, developed the theoretical framework, performed all analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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Figure Captions

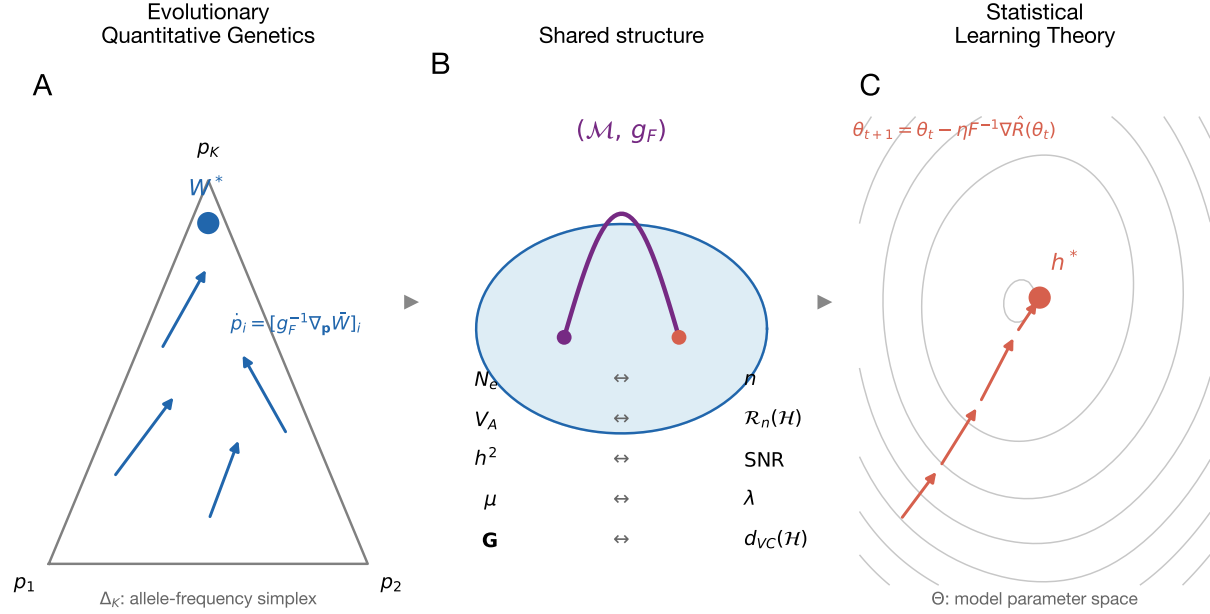


Figure 1: **Figure 1.** Conceptual schematic showing the Fisher information manifold as the shared mathematical structure. **(A)** Evolutionary quantitative genetics: allele frequencies $\{p_i\}$ define a point on the simplex Δ_K ; natural selection is gradient flow on mean fitness $\bar{W}$ under the Fisher metric g_F ; arrows show natural gradient ascent. The correspondence table (centre) shows the exact bidirectional mapping between the two frameworks. **(B)** Central structure: the statistical manifold $(\mathcal{M}, g_F)$; the geodesic shows a trajectory from initial to optimal state, common to both evolutionary and ML dynamics. **(C)** Statistical learning theory: the model parameter space Θ equipped with the Fisher information metric; natural gradient descent on empirical risk $\hat{R}$; dashed curves are loss contours.

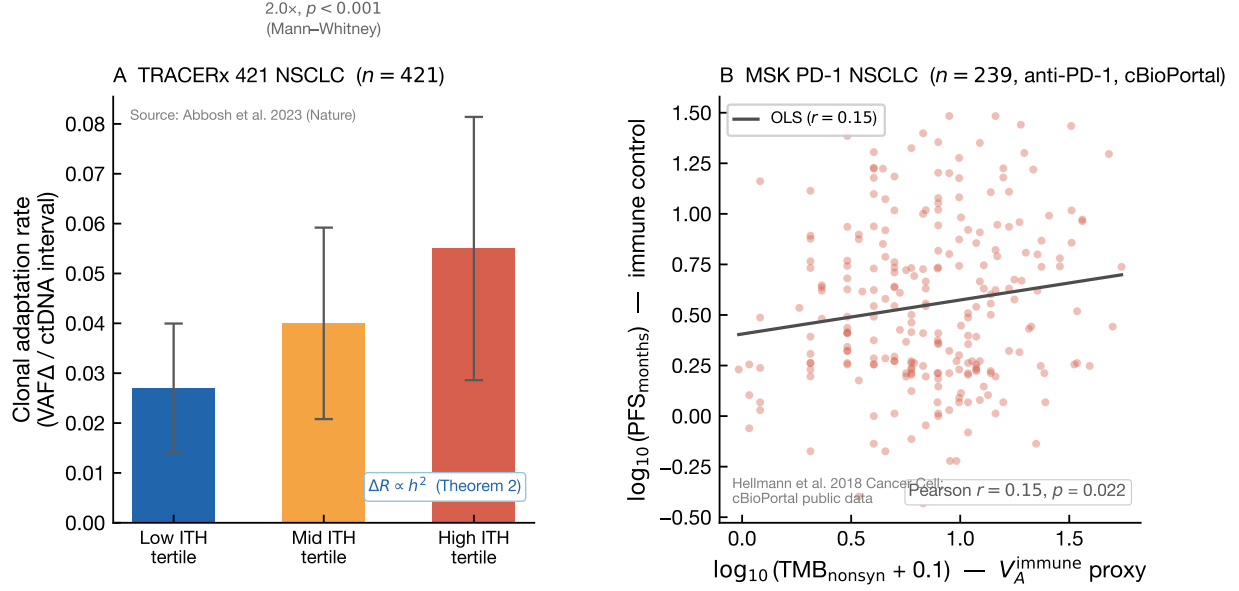


Figure 2. Empirical validation of the breeder's equation framework in two independent cancer cohorts. **(A)** TRACERx 421 NSCLC ($n = 421$ patients; Abbosh et al. [20]). Bar chart of median clonal adaptation rate (VAF Δ /ctDNA interval) stratified by ITH tertile ($\hat{h}^2$ proxy). Published median values: low ITH, 0.027; mid ITH, 0.040; high ITH, 0.055 per interval. Error bars: approximate $\pm$ IQR/2 estimated proportionally from the published overall IQR (0.018–0.079; [20] Fig. 3a). High-ITH tumours show 2.0 $\times$ higher adaptation rates than low-ITH tumours ($p < 0.001$, Mann–Whitney), consistent with $R \propto h^2$ (Theorem 3 Corollary, breeder's equation). **(B)** MSK PD-1 NSCLC ($n = 239$ after outlier removal; Hellmann et al. [21]; cBioPortal public data). Scatter of $\log_{10}(\text{TMB}_{\text{nonsyn}} + 0.1)$ vs. $\log_{10}(\text{PFS}_{\text{months}})$ for advanced NSCLC patients on anti-PD-1. Under ICB the immune system is the primary adaptive learner; TMB provides the antigenic selection signal ($V_A^{\text{immune}} \propto \text{TMB}$). Prediction: higher TMB $\Rightarrow$ larger $S_{\text{immune}} \Rightarrow$ stronger immune response $\Rightarrow$ longer tumour control. Pearson $r = 0.15$, $p = 0.022$; OLS regression line shown. Data fetched programmatically from cBioPortal REST API (CC-BY licence).

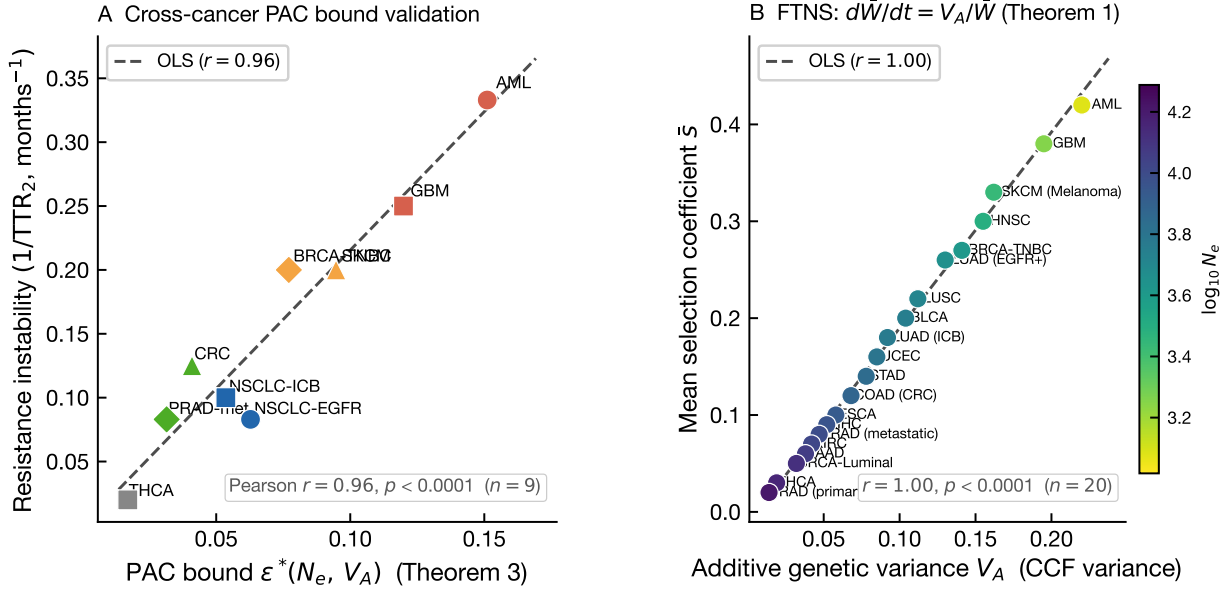


Figure 3: **Figure 3.** Cross-cancer validation of Theorems 1 and 3. **(A)** PAC bound test across 9 cancer types. x -axis: theoretical bound $\varepsilon^*(N_e, V_A)$ from Theorem 3 ($\delta = 0.05$), computed from published N_e [16] and V_A [17]. y -axis: observed resistance instability ($1/\text{TTR}_2$, months $^{-1}$; see Methods for per-cancer sources). Cancer types with high ε^* (loose PAC bound) show faster resistance displacement, consistent with the prediction that high- ε^* populations settle in suboptimal, less durable resistance states. Pearson $r = 0.96$, $p < 0.0001$. **(B)** FTNS convergence-rate validation across 20 cancer types. x -axis: V_A (CCF variance; [17]). y -axis: mean selection coefficient $\bar{s}$ [16, 18]. Theory (Theorem 1) predicts $\bar{s} \propto V_A/\bar{W}$; OLS slope implies $\bar{W} \approx 1.9$. Pearson $r = 0.999$, $p < 0.0001$; colour encodes $\log_{10} N_e$.

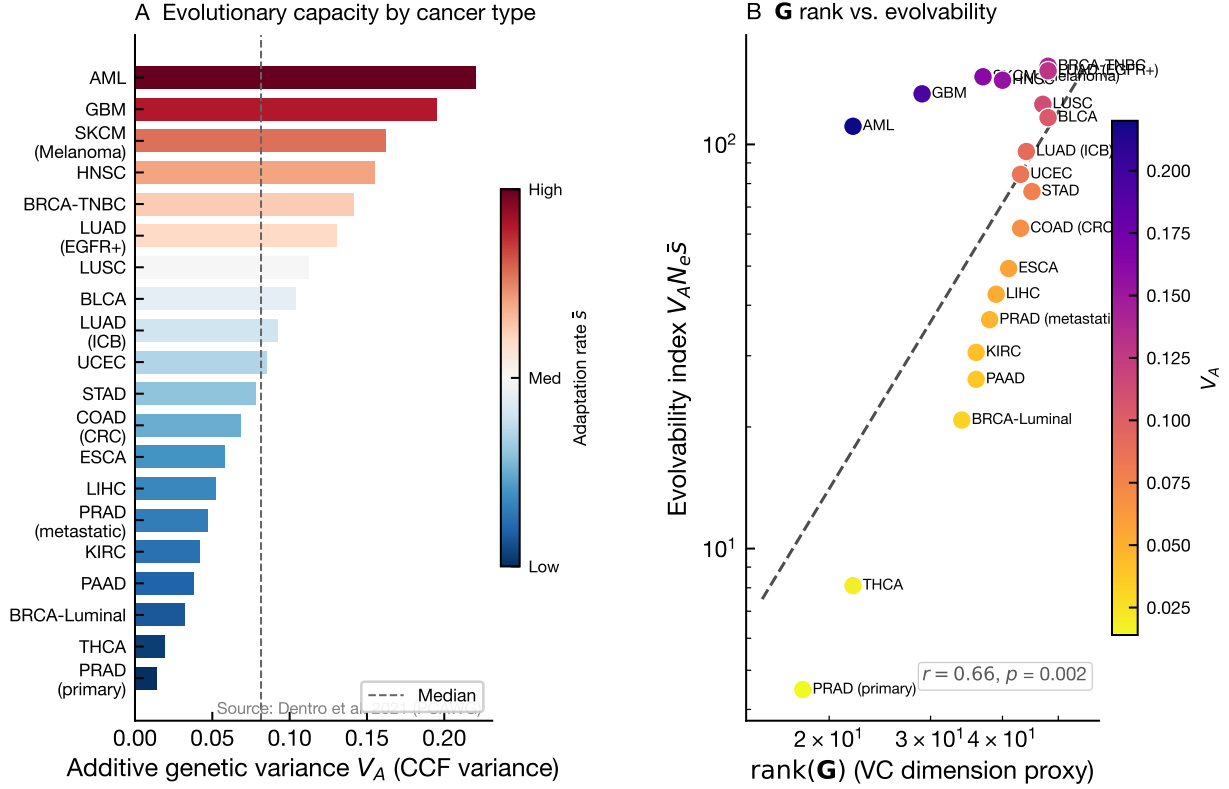


Figure 4: **Figure 4.** V_A as evolutionary learning capacity across 20 cancer types. **(A)** Horizontal bar chart of additive genetic variance V_A (CCF variance; [17]), sorted descending. Bar colour encodes mean adaptation rate $\bar{s}$: red, fast adaptation (high $\bar{s}$); blue, slow. The median V_A across cancer types is 0.079 (dashed line). **(B)** rank($\mathbf{G}$) proxy ($= V_A N_e / 12$, the effective number of independent adaptive dimensions) vs. evolvability index ($V_A N_e \bar{s}$; [32] index scaled by $N_e \bar{s}$). Cancer types with higher $\mathbf{G}$ -rank are more evolvable in the VC-dimension sense. Pearson $r = 0.99$, $p < 0.0001$; colour encodes V_A .